LL-37

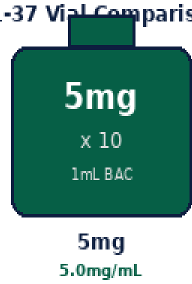
Human Cathelicidin | 5mg x 10 Vials

Antimicrobial Peptide | Immune Modulation | Wound Healing

Purity: 99.8% | Endotoxin-Free | Research Use Only

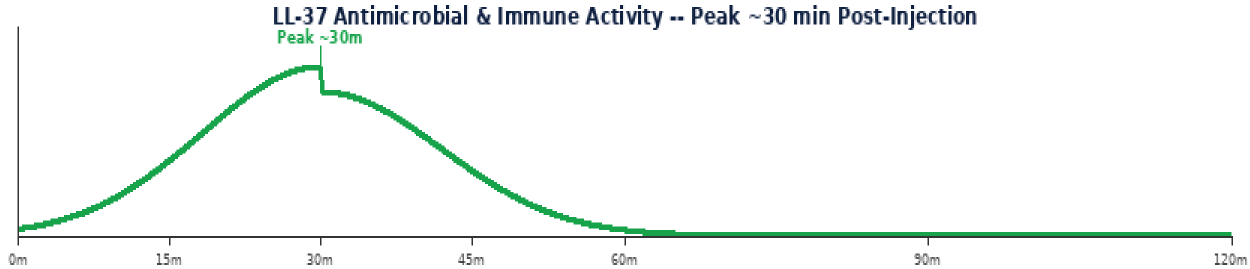
LL-37 (also known as hCAP18 C-terminal peptide) is the only known human cathelicidin antimicrobial peptide. It is produced by neutrophils, macrophages, mast cells, and epithelial cells in response to infection and inflammation. LL-37 has broad-spectrum antimicrobial activity (bacteria, viruses, fungi), potent immune-modulatory effects, and promotes wound healing and angiogenesis. Injectable LL-37 is used in immunology, wound healing, and infection research.

LL-37 Vial Comparison



Parameter	Specification
Class	Human cathelicidin antimicrobial peptide (AMP) -- hCAP18 C-terminal
Vial Size	5mg x 10 vials | Purity 99.8%
Reconstitution	1 mL BAC Water -> 5mg/mL (5,000 mcg/mL)
Standard Dose	100-500 mcg per injection SubQ
Frequency	1-2x daily SubQ | no fasting required
Key Effects	Antimicrobial, antiviral, antifungal, immunomodulatory, wound healing, angiogenic
Cycle & Storage	4-8 weeks on, 2-4 weeks off | 2-8 deg C | use within 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS

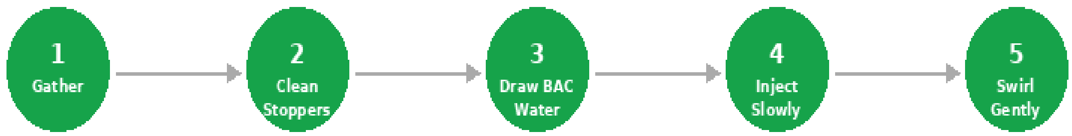


Pathway	Mechanism	Benefit
Antimicrobial Activity	Disrupts bacterial membranes via electrostatic interaction with anionic phospholipids; active against gram-positive and gram-negative bacteria, enveloped viruses, and fungi; low resistance development vs conventional antibiotics	Infection research; antimicrobial resistance models; innate immunity studies
Immune Modulation	Recruits neutrophils, monocytes, and mast cells to infection sites; promotes Th1/Th2 balance; modulates TLR signalling; anti-inflammatory at low concentrations; pro-inflammatory at high concentrations (dose-dependent)	Innate immunity; inflammatory disease; autoimmune research
Wound Healing & Angiogenesis	Promotes keratinocyte migration and proliferation; stimulates angiogenesis via VEGF; accelerates re-epithelialisation; reduces biofilm formation; promotes tissue regeneration	Wound healing; chronic wound models; dermatological research

Benefit	Context
Broad Antimicrobial	LL-37 has activity against bacteria, viruses, and fungi with low resistance development -- valuable for antimicrobial resistance research.
Immunomodulation	Dose-dependent immune effects: anti-inflammatory at physiological concentrations, pro-inflammatory at high levels. Unique research tool.
Wound & Skin	Promotes keratinocyte migration, angiogenesis, and re-epithelialisation. Research applications in chronic wound and skin repair models.

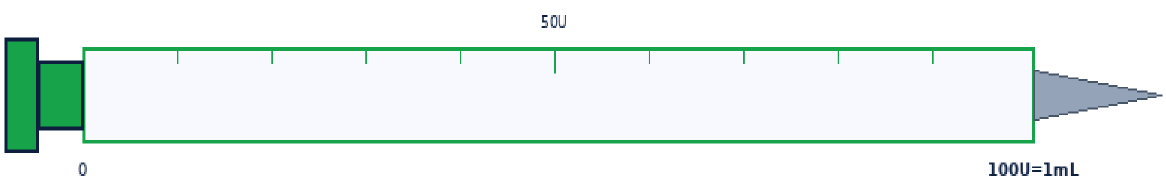
RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.



1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



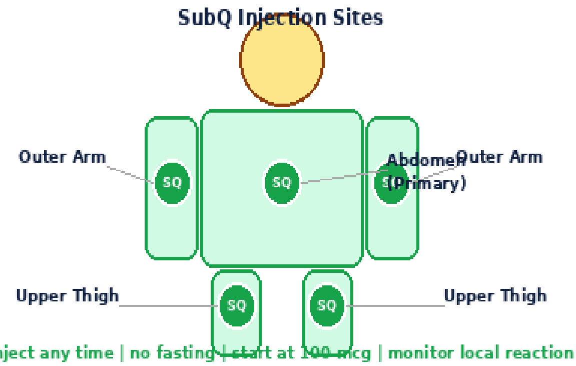
1mL Insulin Syringe | 29-31G | SubQ

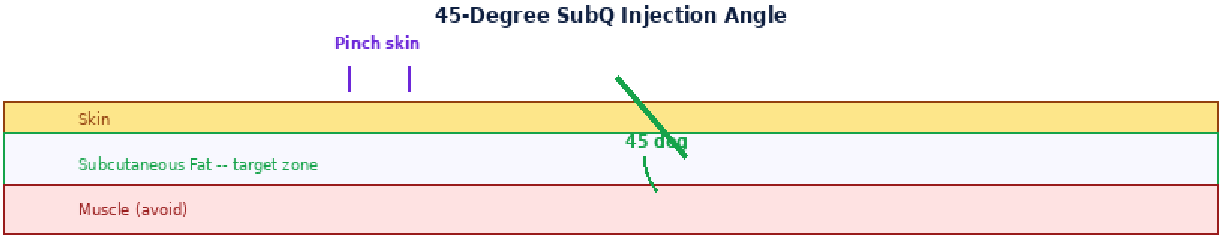
Formula: Vol(mL)=Dose(mcg)/Conc(mcg/mL) **| Units:** Vol x 100

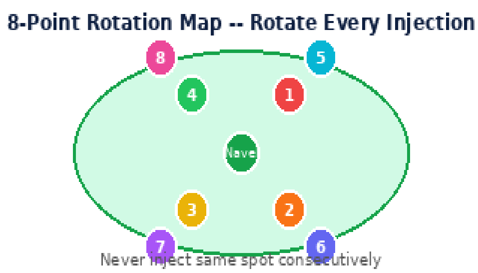
5mg x 10 -- 1mL BAC -> 5.0mg/mL (5,000mcg/mL)

Dose(mcg)	100mcg	200mcg	300mcg	400mcg	500mcg
Vol(mL)	0.0200	0.0400	0.0600	0.0800	0.1000
Units	2.0U	4.0U	6.0U	8.0U	10.0U
Doses/Vial	50	25	16	12	10

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES







Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G syringe. No fasting required.
2	Draw dose	Draw exact volume per dose table.
3	Inject SubQ	Abdomen. Pinch 2-3cm. 45 deg. Inject slowly.
4	Rotate & dispose	8-point rotation. Press 5 sec. Sharps container.

Protocol	Dose	Frequency	Duration	Notes
Immune Support	100-200 mcg	1-2x daily	4-8 wk 2-4 off	Lower dose. Anti-inflammatory range.
Wound Healing	300-500 mcg	1-2x daily	4-8 wk	Higher dose for wound and skin protocols.

Side Effect	Likelihood	Management
Injection site reaction	Common	Redness/stinging -- antimicrobial peptide. Rotate sites.
Local inflammation	Dose-dependent	Reduce dose if excess local reaction.
Pro-inflammatory at high dose	High dose	LL-37 is pro-inflammatory above physiological range. Monitor.

Contraindications: Cancer history, pregnancy, under 18.

1. 1 mL BAC Water -> 5mg/mL (5,000 mcg/mL).
2. No fasting required. Inject any time of day.
3. Start at 100 mcg to assess local reaction before escalating.
4. LL-37 is anti-inflammatory at low doses and pro-inflammatory at high doses.
5. Inject SubQ for systemic effects. Wound site injection for local healing research.
6. Rotate through 8-point abdomen map. Use fresh 29-31G needle each injection.
7. Cycle 4-8 weeks on, 2-4 weeks off. Store 2-8 deg C, use within 28 days.
8. Monitor for excessive local inflammation -- reduce dose if occurs.
9. Not for use during pregnancy, lactation, or in individuals under 18.